Reading Analog Values

To find a solution to this problem, we can write a little code to read the values coming from the microcontroller's analog inputs, using several new functions that are part of the Serial library, itself part of the standard built-in Arduino library. Serial is used by two digital devices to talk to each other and is loosely related to a collection of communication protocols that includes USB. We've actually been using serial communications since the very beginning to upload our source code to the Arduino interface board. Now, using the following code in Listing 6-4, we can use the Serial Monitor included in the Arduino programming environment to actually "see" the values that the ADC is reading.

Listing 6-4. *Analog Serial Monitor*

```
const int analogPin = A0;
int analogValue = 0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  analogValue = analogRead(analogPin);
  Serial.print("analog value = " );
  Serial.println(analogValue, DEC);
  delay(125);
}
```

This little sketch borrows quite a bit from our project sketch from before. It will read the value from analog pin A0 and send that value back to our computer using some of the functions that are a part of the Serial library. In order to see what the Arduino has to say, we need to use the Arduino Serial Monitor to listen to the interface board.

Using the Serial Monitor

The Serial Monitor is accessed through the button at the far right of the toolbar, as shown in Figure 6-7.